



OPTIMIZATION OF AN ACTIVE SUSPENSION

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INTRODUCTION

The vertical dynamics of a vehicle is studied to predict the effect of unevenness on ride comfort and vertical tire contact force. To perform this kind of analysis it is necessary to model the vehicle. It is possible to neglect the driver model, the tire model (except for what concerns vertical dynamics) and advanced suspension models because they are not needed. Therefore, the quarter car model will be used in this report to study the vertical dynamics. The quarter car model is 1D-like model. It includes sprung mass, unsprung mass, a passive suspension represented by spring and damper and the tire represented just by spring.

It is supposed that the vehicle is affected by a bump. The aim is to optimize the suspension to attenuate the undesired effects given by the bump. Then, the suspension can be optimized by means of a PID regulator and an optimizer tool such that the suspension becomes active. The optimizer tool finds the best possible PID parameters. The regulator processes the error signal given by the difference between the reference vertical force and the actual one. The reference vertical force is set to zero to have ideally a perfect and instantaneous damping.

SCHEMES, PARAMETERS AND RESULTS

The quarter car model is implemented in Altair Activate. Especially, two models are created such that only one of them is optimized with a PID regulator and an optimizer tool. At the end, the performances of the two models are compared. The model is shown in Figures 1.1, 1.2, 1.3, 1.4 and 1.5

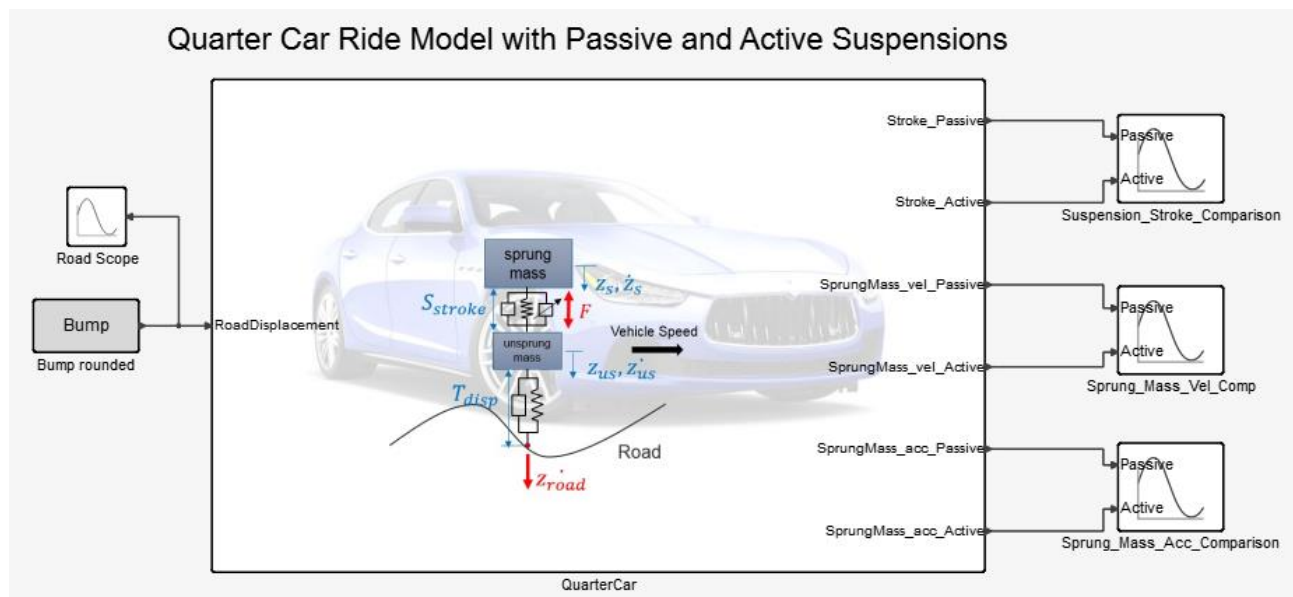


Figure 1.1 Quarter Car Model with Passive and Active Suspensions General Scheme

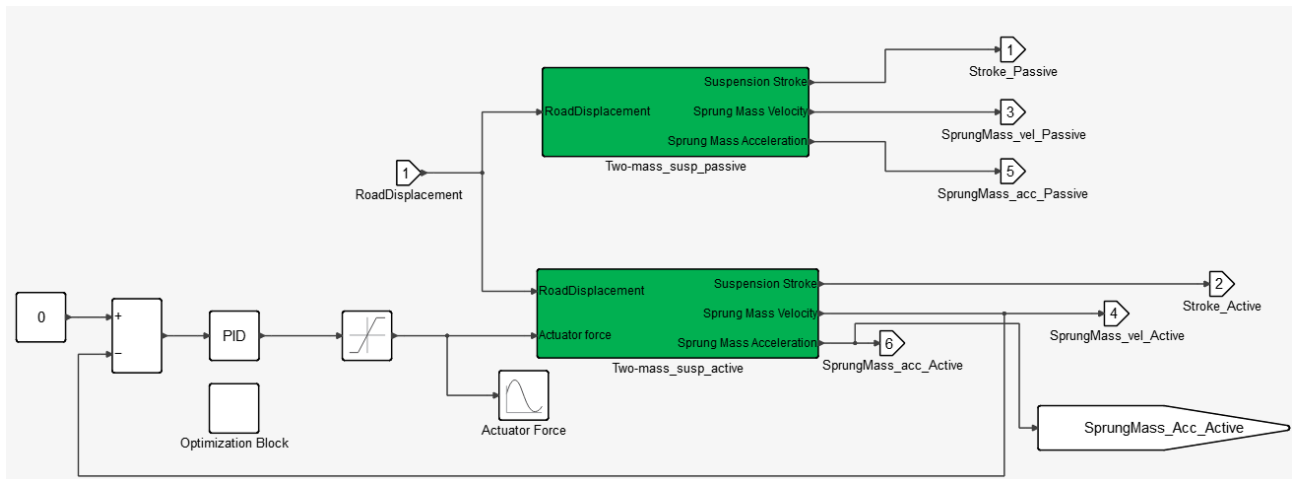


Figure 1.2 Quarter Car Model with Passive and Active Suspensions (inside the subsystem)

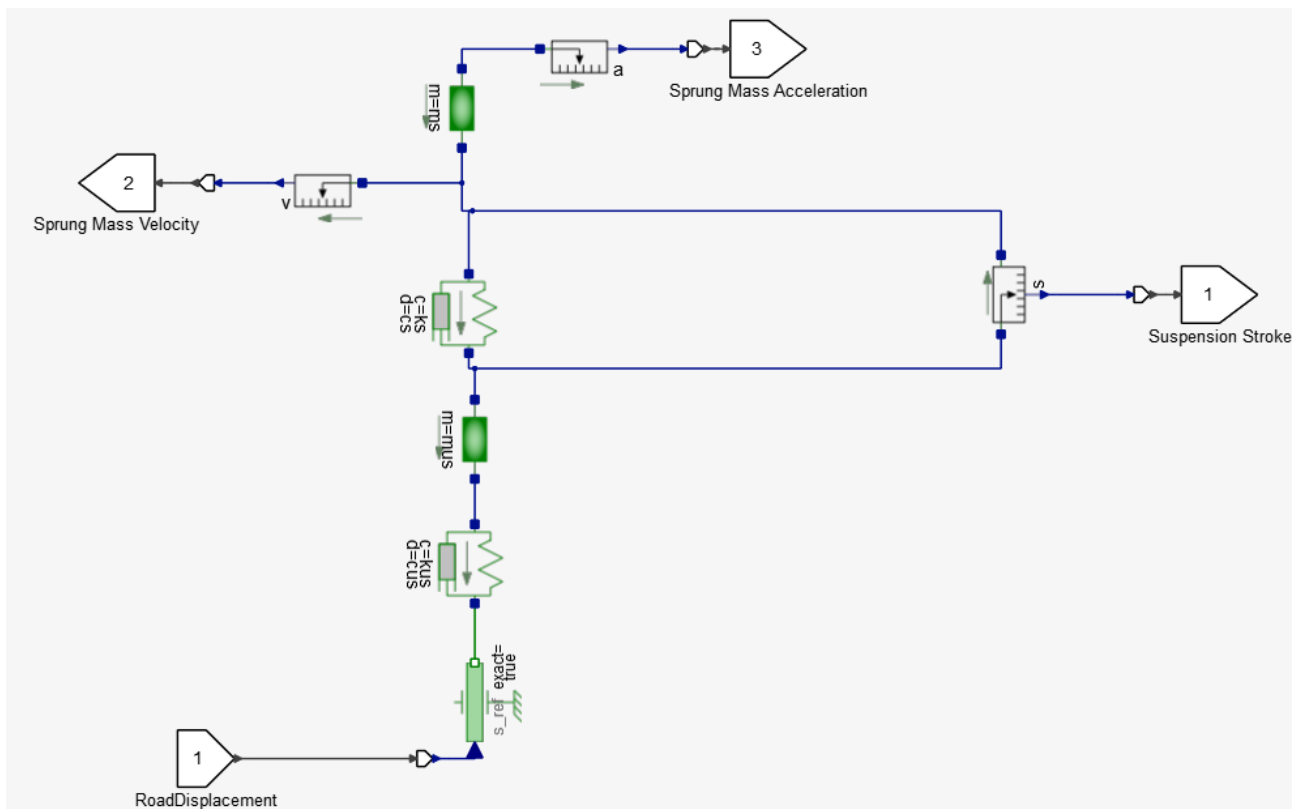


Figure 1.3 Two-mass passive suspension model

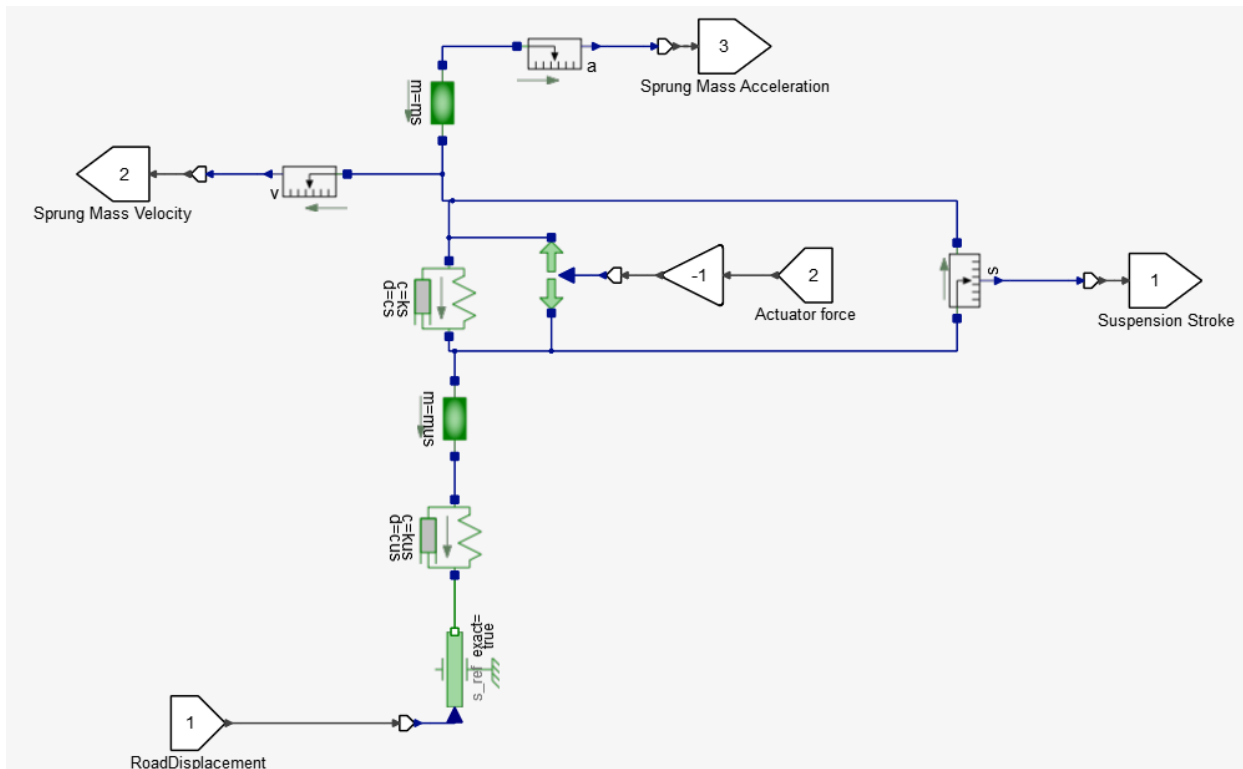


Figure 1.4 Two-mass active suspension model with added force to the vertical quarter car

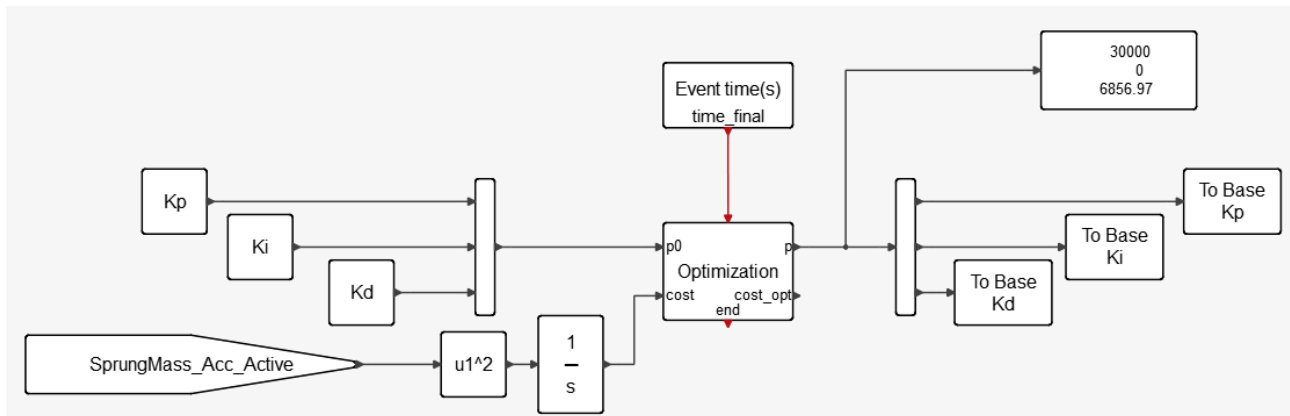


Figure 1.5 Optimization system for the Active Suspension

The model parameters are written in the Table 1:

Table 1 Quarter car model parameters

Parameter	Symbol	Value	Unit
Sprung Mass	ms	476	kg
Unsprung Mass	mus	50	kg
Tire Stiffness	kus	220000	N/m
Suspension Stiffness	ks	26400	N/m
Tire Damping Coefficient	cus	0	Ns/m
Suspension Damping Coefficient	cs	2775	Ns/m

To optimize the active suspension, the PID regulator shown in the Figure 1.2 has been set according to the coefficients calculated by the optimizer tool. At the beginning of the simulation, the PID parameters are null

```
Kp=GetFromBase('Kp', 0);  
Ki=GetFromBase('Ki', 0);  
Kd=GetFromBase('Kd', 0);
```

Figure 2 Code-initialized PID parameters

Then, they are optimized and reported in the Table 2:

Table 2 PID parameters

Parameter	Symbol	Value
Proportional Coefficient	kp	30000
Integral Coefficient	Ki	0
Derivative Coefficient	kd	6856.97

The PID parameters are limited up to 30000 for choice.

Then, the PID response is saturated by the saturation block which limits are ± 2000 .

The optimization system aim is to find the best PID parameters by minimizing the cost function that is given by the integration of the squared sprung mass acceleration.

The vehicle is affected by a bump shown in the Figure 3.1

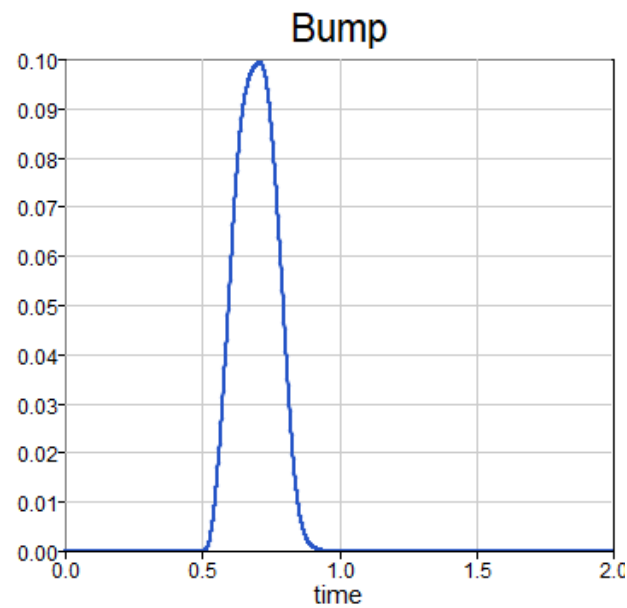


Figure 3.1 Bump evolution

The models responses to the bump are shown in the Figures 3.2, 3.3 and 3.4

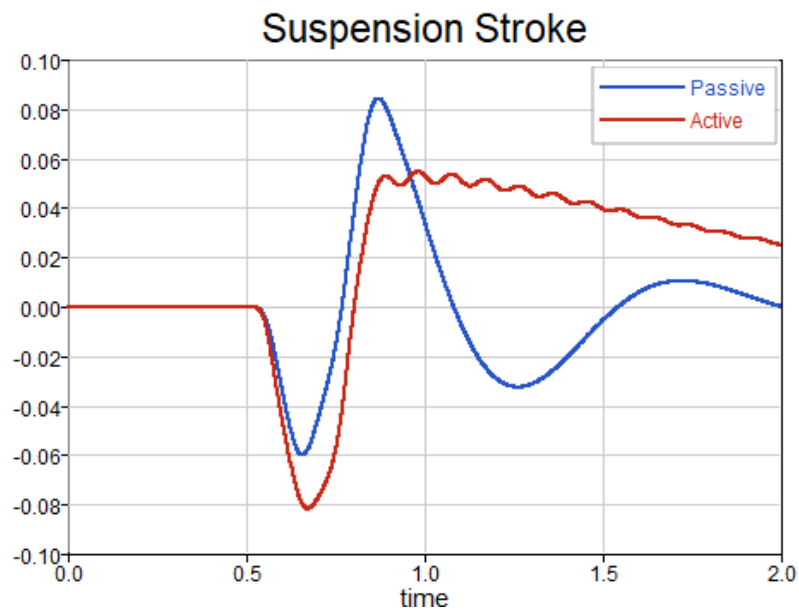


Figure 3.2 Suspension Stroke

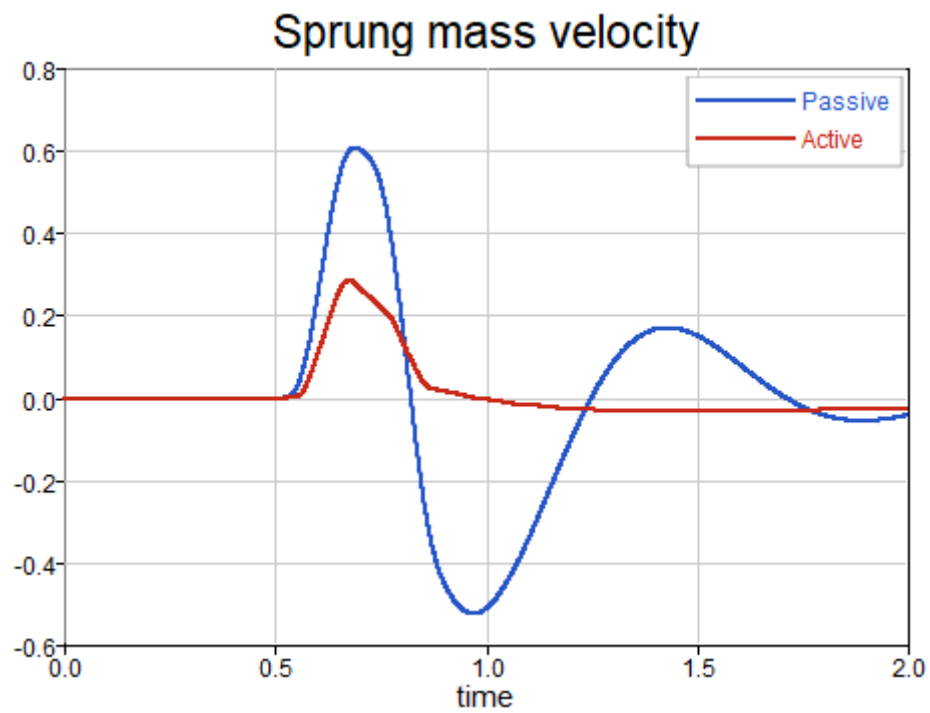


Figure 3.3 Sprung Mass Velocity

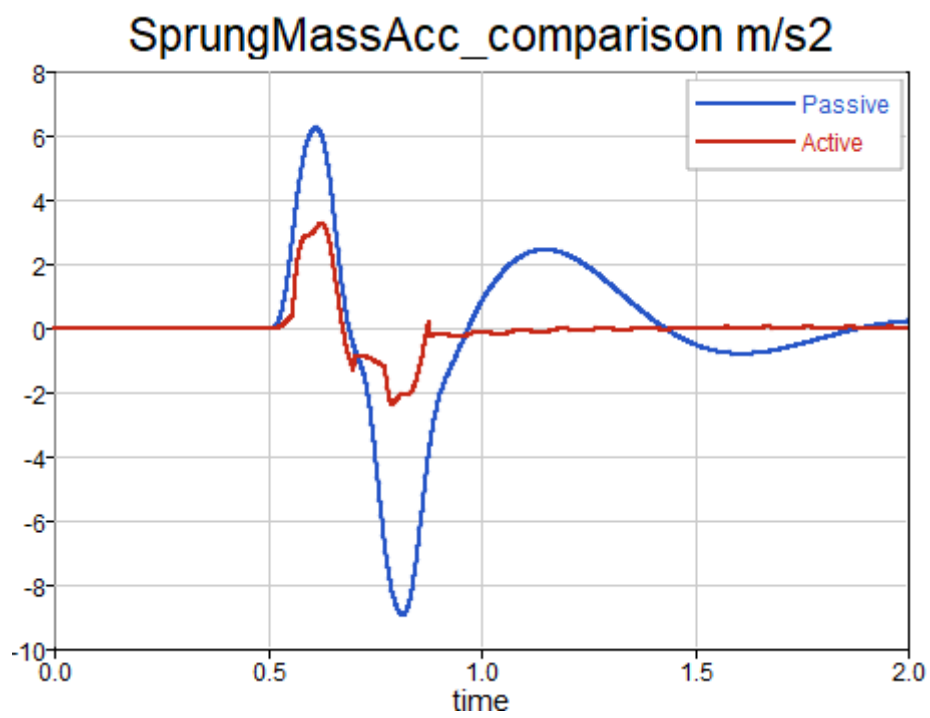


Figure 3.4 Sprung Mass Acceleration

COMMENTS

The suspension stroke for the controlled model is higher (in absolute value) than the one of the non-controlled model in correspondence of the bump action. However, the subsequent response is smoother and slow-varying leading to a better comfort for the passengers, despite the oscillations.

The sprung mass velocity can be detected just during the bump action. Then, it varies around the zero value.

The sprung mass acceleration of the controlled model follows the trend of the one of the non-controlled model but it is damped. The effects of the bump on the acceleration stop after around 0.8 seconds.

Therefore, the optimized active suspension shows better performances than the non-optimized one because the stroke is better attenuated (except during the bump action), and the sprung mass velocity and acceleration are enough damped.